loaded into a hashset in the in-memory cache. And because this is an expensive operation, the cardinality runs as an approximation. So, counts for unique values may not be exact, but they are close.

In addition to the metric aggregations we've discussed, Elasticsearch exposes a few others. Going over all of them in this chapter is not practical, so I strongly recommend that you look at the Elasticsearch documentation to learn about those not covered here.

The next type of metric produces a bucket of documents rather than creating a metric across all documents. These are called *bucket aggregations*, a subject discussed in the next section.

## 13.3 Bucket aggregations

One requirement for data is running grouping operations. Elasticsearch calls these grouping actions *bucket aggregations*. Their sole aim is to categorize data into groups, commonly called *buckets*.

Bucketing is a process of collecting data into interval buckets. For example,

- Grouping runners for a marathon according to their age bracket (21–30, 31–40, 41–50, etc.)
- Categorizing schools based on their inspection ratings (good, outstanding, exceptional)
- Getting the number of new houses constructed each month or year

To play with bucketing aggregations, we reuse a data set we've worked with in the past: the book data. Pick up the data set from GitHub (http://mng.bz/mVZ0) or the book's website, and index it. This sample snippet provides a quick reminder (this is not the full data set):

```
POST _bulk
{"index":{"_index":"books","_id":"1"}}
{"title": "Core Java Volume I â€“ Fundamentals","author": "Cay S. Horstmann",
➥ "edition": 11, "synopsis": "Java reference book that offers a detailed
➥ explanation of various features of Core Java, including exception
➥ handling, interfaces, and lambda expressions. Significant highlights
➥ of the book include simple language, conciseness, and detailed
➥ examples.","amazon_rating": 4.6,"release_date": "2018-08-27",
➥ "tags": ["Programming Languages, Java Programming"]}
{"index":{"_index":"books","_id":"2"}}
{"title": "Effective Java","author": "Joshua Bloch", "edition": 3,"synopsis":
➥  "A must-have book for every Java programmer and Java aspirant,
➥ Effective Java makes up for an excellent complementary read with other
➥ Java books or learning material. The book offers 78 best practices to
➥ follow for making the code better.", "amazon_rating": 4.7,
➥ "release_date": "2017-12-27", "tags": ["Object Oriented Software Design"]}
```

Now that we've primed our server with book data, let's run some common bucketing aggregations. There are at least two dozen aggregations out of the box, each with its own bucketing strategy. As I mentioned earlier, it would be boring and repetitive to document all the aggregations in this book. But once you understand the concept and